

Generative Adversarial Networks for Network Intrusion Detection: Enhancing Cybersecurity Through Synthetic Attack Simulation

Done by: Youssef Bedair, Malak Amr, Mohamed Eisa, Mostafa Hasona, Omar El-hamarawy.

1. Problem Statement

With the increasing sophistication of cyberattacks, traditional Network Intrusion Detection Systems (NIDS) struggle to detect zero-day attacks due to their reliance on known attack patterns. Current machine learning-based NIDS models require large labeled datasets, which are often limited in diversity and do not adapt well to evolving threats. There is a need for a system that can continuously learn and generate realistic cyberattacks to improve IDS performance.

Comparison Table: Old vs GAN-Based Solution			
Feature	Signature-Based	Traditional ML	GAN-Based NIDS
Zero-Day Attack Detection	No	Limited	Yes
False Positive	Low	High	Reduced
Self-Learning Ability	No	Limited	Yes
Dependancy on Labeled Data	High	High	Lower
Adaptability to New Threats	No	Limited	High
Performance on Encrypted Traffic	Poor	Limited	Good

2. Objectives

- Develop a GAN-based model capable of generating synthetic attack traffic.
- Train an IDS to detect real and GAN-generated attacks.
- Evaluate the IDS's performance in detecting zero-day threats.
- Deploy a real-time monitoring system for cybersecurity defense.

3. Scope of Work

- **Functional Aspects:** GAN model development, IDS training, real-time monitoring.
- **Technical Aspects:** Data preprocessing, model training, feature extraction, IDS evaluation.
- **Users & Applications:** Cybersecurity analysts, enterprise networks, cloud security providers.

4. Domain-Specific Constraints & Challenges

- **Data Availability:** Access to high-quality network traffic data.
- **Computational Requirements:** Training deep learning models requires significant resources.

- **Adversarial Robustness:** Ensuring the generated attack patterns are diverse and realistic.

5. Chosen Technologies

- **GAN Framework:** PyTorch/TensorFlow for model development.
- **IDS Algorithms:** Autoencoders, LSTMs, Transformer-based models.
- **Data Processing:** Pandas, Scikit-learn.
- **Deployment:** Snort/Suricata for real-time monitoring, FastAPI for integration.

6. Available Solutions and Added Value

- **Traditional IDS** (Signature-based, anomaly detection) lacks adaptability to new attacks.
- **Machine Learning IDS** requires large labeled datasets.
- **Proposed GAN-based IDS** enhances adaptability by generating realistic attack traffic, allowing IDS to train on evolving threats.

7. Methodology

- **Data Collection:** Gather real network traffic datasets (CICIDS2017, NSL-KDD).
- **Preprocessing:** Normalize, encode, and extract relevant features.
- **GAN Training:** Train a generator to simulate attacks and a discriminator to classify real vs. synthetic traffic.
- **IDS Training:** Train a machine learning-based IDS on a combination of real and GAN-generated data.
- **Evaluation:** Test IDS performance against real-world attack patterns.
- **Deployment:** Integrate IDS into a real-time monitoring system.

8. Expected Outcomes

- A GAN model capable of generating high-fidelity synthetic cyberattacks.
- An IDS that detects zero-day attacks with higher accuracy.
- A real-time intrusion detection system that adapts to evolving threats.
- Improved cybersecurity resilience for enterprise and cloud-based networks.

9. Timeline & Project Duration

Phase	Task	Duration
1	Research & Data Collection	1 Week
2	Model Implementation	2 Weeks
3	Optimization & Testing	1 Week
4	Deployment & Evaluation	2 Weeks

11. References

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